



Report

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2026 / 09 / 04

Summary

① Tasks Status

② Results

③ Selected Article

General Tasks Status

Task	Status	
Correlation Inference Study - Classical Approach		
Correlation Inference Study - Quantum Learning Approach		
2026-11 Paper		
2026-12 LASNPA Presentation		

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Uncertainty correlation effect	✓	Plots
RHIC Au-Au/d-Au Studies JETSCAPE		
↪ Uncertainty correlation effect	✓	Plots
Atlas ins1759506 studies		
↪ Uncertainty correlation inference	🕒	Notebook
Alice PbPb2760 Scaled Spectra studies		
↪ Uncertainty correlation inference	🕒	Notebook
Synthetic data studies		
↪ Uncertainty correlation inference	🕒	Notebook

Atlas ins1759506 studies

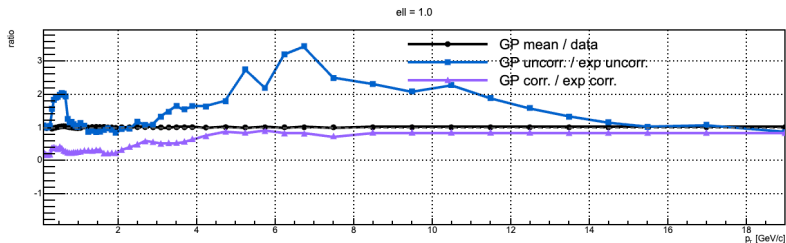


Figure: GP fit: fixed ell = 1.0

Atlas ins1759506 studies

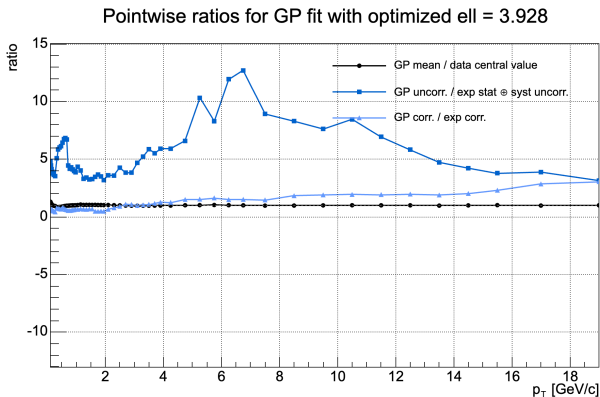


Figure: GP fit: optimized ell

Atlas ins1759506 studies

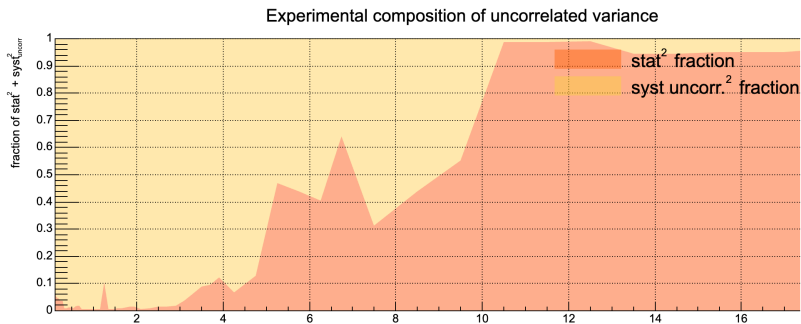


Figure: Ratio between white noise and systematic uncorrelated noise

Alice PbPb2760 Scaled Spectra studies

ALICE Pb-Pb 2.76 TeV Grad, 0-5%

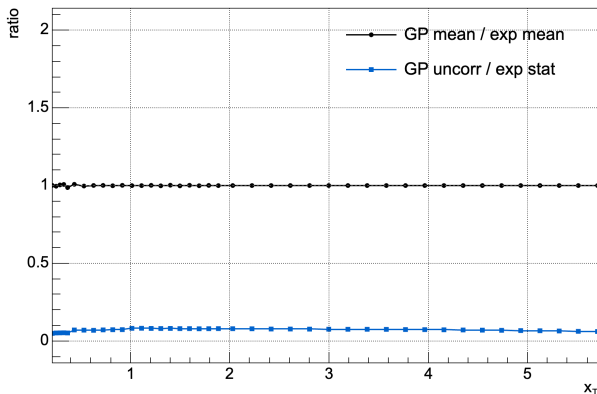


Figure: GP fit: fixed ell = 0.2

Alice PbPb2760 Scaled Spectra studies

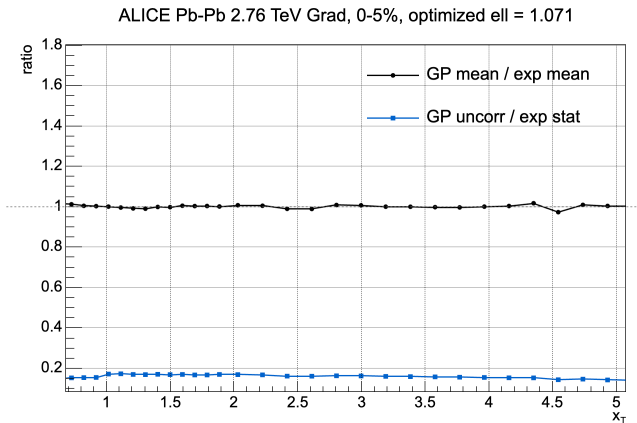


Figure: GP fit: optimized ell

Selected Article

Responses of multiparticle observables to multidimensional nuclear deformation in relativistic heavy-ion collisions

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Relativistic heavy-ion collisions provide a unique opportunity to probe ground-state nuclear structure through its imprint on the initial collision geometry. We investigate how multiparticle observables respond to combined variations of quadrupole deformation, triaxiality, and hexadecapole deformation, using $^{129}\text{Xe}+^{129}\text{Xe}$ collisions as a representative testing ground. We perform a joint analysis in the three-dimensional $(\beta_2, \gamma, \beta_4)$ parameter space and construct initial-state estimators for several flow and mean transverse momentum correlation observables. At the initial-state level, ρ_2 is primarily sensitive to β_2 and γ , with its sensitivity to γ enhanced at nonzero β_2 . The nonlinear response coefficient $\chi_{4,22}$ is predominantly sensitive to β_4 , while its dependence on β_2 and γ remains comparatively weak. Higher-order correlators exhibit more complex multidimensional response patterns; in particular, ρ_{224} shows a dependence on γ and β_4 that becomes more pronounced at finite β_2 . We further employ the iEBE-VISHNU hybrid model to examine whether these deformation sensitivities survive the subsequent dynamical evolution. The final-state calculations indicate that the sensitivity of ρ_2 to β_2 and γ is largely preserved, whereas the β_4 sensitivity of $\chi_{4,22}$ is substantially reduced. For the other higher-order observables, the initial-state sensitivities are modified by the evolution or cannot be resolved with the present statistics.

Figure: Multidimensional nuclear structure deformation study.